

## PHYS 627/ECE 523 : Quantum Information Science

Professor: Dr. Natalie Klco (natalie.klco@duke.edu)

Office: 251 Physics or DQC 4308

Office Hours: By Appointment

TA: Bahaa Harraz

### Class Location and Hours:

Dates: 01/07/2026 - 04/15/2026

Time: Monday and Wednesday, 10:05 am to 11:20 pm

Location: Wilkinson 136

Instruction Mode: In Person

### Outcomes:

The goal of this course is to develop practical skills in the analytic and numerical exploration of quantum information processing protocols. You will be expected to demonstrate mastery through classical simulation and mathematical/physical reasoning of quantum circuit logic and performance efficiency. Naturally, exercises are cumulative as capabilities and algorithmic subroutines are incorporated into your knowledge and personal simulation environment(s)<sup>1</sup>. The primary objective of this course is development of necessary skills—intuition, understanding, perspective, resources and technical skills—to substantially engage with the quantum computing and quantum information literature and community.

### Course Texts and Resources:

- P. Kaye, R. Laflamme, and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press, 2007. <https://find.library.duke.edu/catalog/DUKE99119520109108501>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000. <https://find.library.duke.edu/catalog/DUKE99119504398108501>
- John Preskill's Physics 219 lecture notes on Quantum Computation <https://www.preskill.caltech.edu/ph229/>
- arXiv <https://arxiv.org/>
- Inspire <http://inspirehep.net/>
- Google Scholar <https://scholar.google.com/>
- Duke off-campus proxy <https://library.duke.edu/research/library-access-button>

Collaboration with peers is also a resource. The class will have a Slack channel to facilitate my communication with you as well as your communication with each other.

### Grading and Assessment:

The default grading will be calculated as:

- Exercise Submissions: 85%
- Project/Presentation: 15%

As many of you are already or may consider pursuing careers in the broad ecosystem of quantum information science, it is anticipated to be more valuable for the development of your research

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<sup>1</sup>Coding language of your choice, e.g., mathematica, python, c++. Specialized quantum simulators may also be utilized e.g, Qiskit, Cirq. If the latter is chosen, understanding of the internal workings of provided functions must be clearly demonstrated prior to their use.

trajectories to present integrated understanding through projects synthesizing course material than performance in a written exam. The course design reflects this perspective.

Full credit will be granted for completion of  $\sim 80\%$  of the homework exercises. In the grading, no partial credit will be given. However, solutions may be submitted a number of times limited only by the patience of the TA. A complete solution is expected to be **pedagogically crafted to demonstrate thought process**. If collaborating with others, contribution acknowledgments must be noted.

- Homeworks will be expected not only to contain the information of correct solutions, but also to convey understanding. This is a skill that will serve you well both in guiding your own research trajectories as well as in writing your own research articles. Just as no publishable research article is a series of mathematical equations without explanation, your submissions are expected to be contextualized and presented **with written narration**. Such scientific narration may, for example, explain the physical intuition associated with sections of the calculation, detail the mathematical justification for intermediate steps, process and reflect upon experimental connections, or evaluate limiting cases as “sanity checks” that the solutions are consistent with basic physical expectations.
- Science is an interactive and collaborative endeavor that maintains individual metrics and assessment criteria and respects intellectual property. Documenting appropriate attribution for the ideas presented in your work is an essential skill of a responsible scientist. As such, homework assignments should be thoroughly cited. Collaboratively developed solutions, or those informed by a text or online resource, should be accompanied by citations or acknowledgements. All uncited content of your submission will be assumed to be original material.

### Class Structure:

Class time will be approximately equally distributed between lecture and discussion/collaboration time. Discussions may be focused upon prior lecture material, progress on problem sets or projects, or reflections from recent reading. Exact distribution between these class modes will vary according to material and optimized value to student development.

### Course Schedule:

The following is a baseline schedule. Modifications will be made in response to student interactions. We will discuss several options for the project/presentation in early March.

| Date              | Note                                | Topic                                   |
|-------------------|-------------------------------------|-----------------------------------------|
| January 7, 2026   |                                     | Interferometry and Computation          |
| January 12, 2026  |                                     | Visualizing Gates and Noise             |
| January 14, 2026  |                                     | Collaboration Session I                 |
| January 19, 2026  | Martin Luther King, Jr. Day holiday |                                         |
| January 21, 2023  | <i>Submission Pulse</i>             | Arbitrary Basis Measurements            |
| January 26, 2026  |                                     | Parity and Non-Destructive Measurements |
| January 28, 2026  |                                     | Collaboration Session II                |
| February 2, 2026  |                                     | Modeling Noisy Environments             |
| February 4, 2026  | <i>Submission Pulse</i>             | Collaboration Session III               |
| February 9, 2026  |                                     | Entanglement and Bell's Inequalities    |
| February 11, 2026 |                                     | Collaboration Session IV                |
| February 16, 2026 |                                     | Identifying Entanglement                |
| February 18, 2026 | <i>Submission Pulse</i>             | Collaboration Session V                 |
| February 23, 2026 |                                     | Entanglement for Information Processing |
| February 25, 2026 |                                     | Collaboration Session VI                |
| March 2, 2026     |                                     | Noisy Quantum Sensing and Teleportation |

|                |                         |                                              |
|----------------|-------------------------|----------------------------------------------|
| March 4, 2026  | <i>Submission Pulse</i> | Collaboration Session VII                    |
| March 9, 2026  | Spring Recess           |                                              |
| March 11, 2026 | Spring Recess           |                                              |
| March 16, 2026 |                         | Circuit Rudiments and Hamiltonian Simulation |
| March 18, 2026 | <i>Submission Pulse</i> | Collaboration Session VIII                   |
| March 23, 2026 |                         | Positive Operator-Valued Measurements        |
| March 25, 2026 |                         | Collaboration Session IX                     |
| March 30, 2026 |                         | Mixed-state Entanglement and Majorization    |
| April 1, 2026  | <i>Submission Pulse</i> | Collaboration Session X                      |
| April 6, 2026  |                         | Student Presentations                        |
| April 8, 2026  |                         | Collaboration Session XI                     |
| April 13, 2026 |                         | Student Presentations                        |
| April 15, 2026 | <i>Submission Pulse</i> | Student Presentations                        |
| April 20, 2026 | Graduate Reading Period | Asynchronous Collaborations                  |
| April 22, 2026 | Graduate Reading Period | Asynchronous Collaborations                  |
| April 27, 2026 | Final Examinations      | Asynchronous Collaborations                  |
| April 29, 2026 | Final Examinations      |                                              |

### **Code of Conduct and Academic Integrity:**

All students must adhere to the Duke Community Standard (DCS): Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, and accountability. Citizens of this community commit to reflect upon these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity. To uphold the Duke Community Standard, students agree:

- I will not lie, cheat, or steal in my academic endeavors;
- I will conduct myself honorably in all my endeavors;
- I will act if the Standard is compromised.

Behaving with **integrity and respect** is part of our responsibility to our shared learning community. Acts of academic misconduct includes plagiarism (representing the work of others as your own, i.e., without appropriate attribution to the original author(s)).

Violations of the DCS may be considered grounds for dismissal from the class.

### **Religious Accommodations:**

Students are permitted by university policy to be absent from class to observe a religious holiday. Please notify the instructor.

**Illness:** Please do not come to class if you are likely contagious.

### **Inclement Weather Policy:**

In the event of inclement weather or other connectivity-inhibiting events, modifications to the schedule will be communicated and asynchronous alternatives may be announced.